a Validation targets support different claims

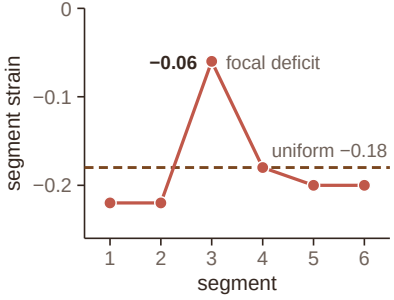
● Validation target	✓ Direct claim it supports	✕ Does not establish on its own
Known motion phantom, STRAUS, MRXCAT2.0	technical error in the simulated regime; capacity needs direct field fitting	in-vivo accuracy
Paired DENSE or tagging same session, registered	material-sensitive agreement under matched definitions	focal location or extent, unless measured
Scan–rescan same subject, re-imaged	precision; accuracy needs a reference	bias
LGE or tissue characterisation scar or fibrosis label	biological association; motion error stays unmeasured	pointwise displacement error
Outcome events, prognosis	prognostic association; utility needs a decision study	decision impact

b Three questions, complementary rather than sequential

Question	Targets in a	Resources (Table 4)	
Can it recover known motion? exact truth, simulated domain	Known motion	STRAUS MRXCAT2.0 CMAC	**3 templates × 6 states** **programmable generator** **dynamic phantom**
Does it agree in vivo, and repeatably? material-sensitive reference; precision	Paired DENSE or tagging Scan–rescan	Stanford CMAC	**55; pairs unverified** **15 volunteers, 12 landmarks**
Does it matter biologically or clinically? tissue state, prognosis, decisions	LGE or tissue Outcome	Clinical cohorts	**LGE or outcomes**

The combination follows the intended use and a pre-specified measurement contract.
Counts are resource units, not independent test participants.

c Same mean, focal deficit



segments 1–3, focal	**−0.22 · −0.22 · −0.06**
segments 4–6, focal	**−0.18 · −0.20 · −0.20**
global mean, both fields	**−0.18**
range, focal / uniform	**0.16 / 0**

Recovering the mean does not establish deficit recovery.